

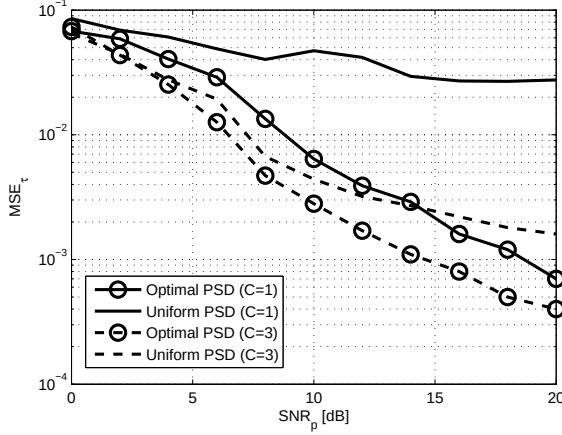
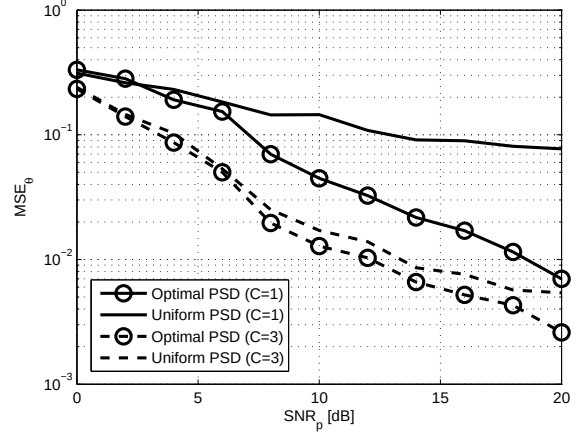
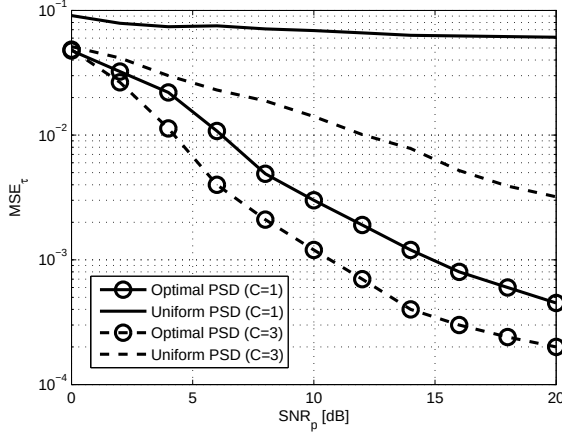
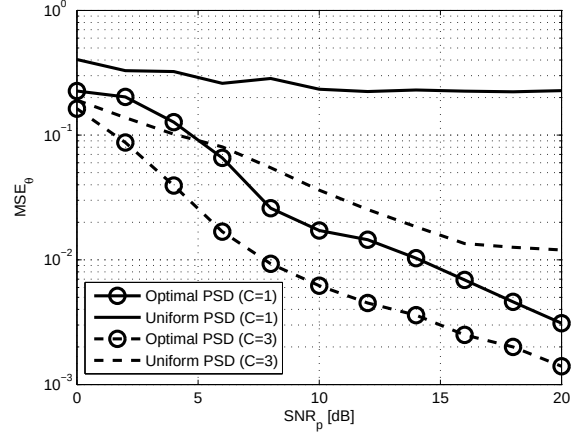
(a) $F = 1$, MSE of timing offset(b) $F = 1$, MSE of phase offset(c) $F = 2$, MSE of timing offset(d) $F = 2$, MSE of phase offset

Fig. 3. MSE for joint phase and timing estimation (31) versus the SNR_p with $A = 0.7$, $N = 100$ and $N_p = 16$.

$k/N_p(n + \tilde{\tau})$. We evaluate the performance of the estimator (31) in terms of MSEs of time and phase offsets by considering the quantization noise with both the optimized PSD obtained from Algorithm 1 and a white PSD that is constant across all frequencies and is selected to satisfy the from the constraint (24b). The white-PSD compression scheme is considered as reference as it does not attempt to optimize quantization with the aim of enhancing synchronization.

Fig. 3(a) and Fig. 3(b) illustrate the MSE of the timing and phase offset estimates, respectively, as a function of SNR_p for $C = 1$ bits/sample and $C = 3$ bits/sample with $F = 1$, $A = 0.7$, $N = 100$, and $N_p = 16$. In addition, we plot the MSE of the timing and phase offset estimates in case of $F = 2$ in Fig. 3(c) and Fig. 3(d), respectively, under the same parameters. We observe